Horizon-dependent Performance and Undercoverage in Probabilistic Movement Forecasting

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Abstract

We study the relation between distributional score, position accuracy and uncertainty coverage for fitted stochastic movement predictors. Evaluation uses 46 unique recording-hash blocks selected retrospectively for temporal and future-route map support, five fixed simulation seeds and 512 particles. Horizons are nominal 1, 5, 15 and 30 minutes in the saved clock; physical clock provenance and participant independence are not established, and this is not a previously untouched evaluation cohort. Full's equally weighted energy score (ES) is 477.85 m, versus 824.73 m for deterministic constant-speed extrapolation. The inertial reference is better at 1 and 5 minutes and Full at 15 and 30 minutes. Full's nominal 90% marginal prediction disk covers only 46.52% of 30-minute targets. The original 21 primary-method comparisons yield 14 practically equivalent and seven inconclusive finite-algorithm states; none establishes an effect wholly beyond the registered practical margin. Resolution-invariant states remain inconclusive. Terrain-family inference is unavailable because required forecasts failed; this does not establish that terrain is ineffective. These conditional findings motivate joint reporting of score, point error, coverage and evidence availability, not an operational reliability claim or general ranking of algorithms.

Evidence version. The once-only saved-output audit and the fixed-budget public aggregate replay are complete. This revision reorganizes their results without changing forecasts, fits or statistical rules. The scientific supplement supplies full scientific specifications and tables; the audit record supplies version, execution and review history. Authorship, route-publication permission and human acceptance remain separate decisions.

1 Introduction and research scope

A useful movement forecast must describe plausible future positions without hiding uncertainty. A low distributional score, a small position error and a well-calibrated prediction region answer different questions. They need not improve together. We examine these distinctions in a limited collection of interpretable fitted stochastic predictors, rather than proposing a new general stochastic differential equation (SDE) theory.

Continuous-time correlated random walks model irregularly observed movement and persistence [1]. Switching state-space models represent latent regimes [2]; their regime identities require care when parameters are transferred between batches. Step-selection models link movement and environmental selection [3, 4], but that relation cannot be inferred from arbitrary unbalanced feature ablations. Learned trajectory models [5, 6] and neural SDE path generators [7] provide broader context, not executed performance baselines for this study. Interpretable stochastic system identification [8] motivates explicit drift and noise specifications. Our contribution is a bounded empirical examination of horizon-specific trade-offs and undercoverage in the predictors actually evaluated.

Proper distributional scoring [9, 10] rewards a predictive distribution relative to realized targets, but does not certify that a particular marginal region attains its nominal coverage. Consequently we keep ES, error of the ensemble mean, and disk coverage/size separate. The simple inertial reference establishes the short-versus-long-horizon reversal; internal comparisons then identify which implemented procedures change these diagnostics. They do not establish the necessity of a complex random model relative to a simple probabilistic motion baseline.

The original registered questions remain traceable: RQ1 asks about method component comparisons; RQ2 asks about terrain increments beyond the history baseline. RQ1 now has conditional fixed-algorithm results, with several comparisons changing multiple factors or sharing parameters. RQ2 has a design and an explicit unavailable result, not a positive or null terrain answer. The paper is organized around what this evidence can actually show.

2 Predictors and what the comparisons change

2.1 Ordinary Full and its actual inputs

Let $X = (X_e, X_n)^\top$ denote position in the saved local coordinate frame (metres), and t the saved time coordinate (seconds). The origin is the last permitted prefix position. The first visible position anchors the frame. Full uses three fitted modes and the position dynamics

$$dX_t = \Theta_{m_t}^\top \psi(X_t, t)\, dt + L_{m_t}\, dW_t, \quad \psi = (1, X_e, X_n, S)^\top, \quad L_m L_m^\top = Q_m = \tau R_m. \tag{1}$$

Here Θ_m gives drift in m/s, S is the solar/calendar input computed from the retained absolute tick [15], and R_m is a fitted rate-residual covariance in $\mathrm{m}^2/\mathrm{s}^2$. Its conversion to Q_m uses the model observation interval τ, not the integration step h. The physical interpretation of seconds and daylight remains conditional on clock provenance (Section 3). Saved mixing weights π_m come from fitting/adaptation, not a posterior update from a new visible prefix. Each particle draws a mode from saved π at the origin. Segment-fixed Full retains that draw; only the switching routes redraw at their τ events, not at every integration substep or scoring output.

Table 1: Actual prefix information. A buffer being updated is not proof that the drift reads it. The three routes are not interchangeable.

Route	Drift inputs and interpretation
Ordinary Full	Position, calendar proxy and saved π; no initial speed or signed recent heading at fixed frame, position, clock and random stream.
Explicit features	Speed norm and position-radial direction; also changes three modes to one. Radial direction is not velocity direction.
Terrain base/full	Secant velocity and signed three-position history, with retained map features in all-terrain.

Changing a prefix can change the frame anchor, the numerical position and origin-specific random-stream identity. Full therefore does not "ignore the prefix". However, after these inputs are held fixed, changing only initial velocity leaves ordinary Full unchanged. On the explicit route, reversing velocity at the same norm does not change its speed feature; the position-radial direction has its own zero-position convention.

Each training/adaptation/validation batch is independently sorted by segment net-displacement heading and split into three rank groups. Mode labels 0/1/2 in code correspond to mathematical

modes 1/2/3. Training has 110/109/109 segments, adaptation 26/25/25, and validation 27/27/27; these are segment counts, not transition frequencies or π_m. Reptile tasks sort their own batches. Blending uses matching indices without a semantic matching step. Missing per-mode direction support and intermediate parameter snapshots prevent a claim that these indices represent matched behaviours. Thus transfer results concern this sorting-and-interpolation procedure, not meta-learning or adaptation in general. Full tie/branch-cut rules and the original 84 groups, 30 eligible tasks and 258 task segments are retained in the scientific supplement.

2.2 Fitting, calibration and propagation

Base drift fitting uses ridge regression on displacement rates. Ordinary residual covariance is centred with denominator $n_m - 1$ and a $10^{-8}I$ jitter before the saved scale and $Q = \tau R$ conversion. Full calibrates a finite scale grid while fixing drift-blend coefficient $\eta = 0$. Mixed/pure-ES additionally fit validation drift and choose η, s on a finite grid, whose largest drift blend is 0.5. These are calibration programs, not end-to-end ES training or global optima.

The selection proxy uses one-step rate residuals and a single Gaussian with the unweighted arithmetic covariance $M^{-1}\sum_m R_m$, not the π-weighted multi-mode, multi-step position distribution eventually scored. GMM first fits a pooled drift, groups its residuals once by $\arg\max(r_e, r_n, -r_e - r_n)$, then refits group-specific drifts. Validation residuals instead select components using segment-heading mode labels. This is a real label mismatch, not a likelihood-EM fit. The proxy/task mismatch is a plausible explanation for undercoverage, but the current results do not identify it as the cause.

The registered integration cap is $h = 5$ saved seconds, with $N = 512$ particles and five fixed seeds. FP denotes conditional Gaussian moment propagation evaluating nonlinear conditions at the conditional mean; MC evaluates them along individual paths. Local affine substeps may be exact, but this is not an exact solution of a globally nonlinear SDE or global Fokker–Planck equation [11]. The four targets are generated from a continuing joint stochastic path, not four independently sampled marginals. Common random indices require the same grid, origin, pair and seed; they neither prove variance reduction nor Brownian consistency across different step grids. All original propagation and numerical-qualification limits are retained in the scientific supplement.

Parameter equality, array equality, score equality and random-stream equality are distinct claims. Full01/07/11/18/20 and the additional Full reference remain separately named; their saved scores are never pooled to make an artificial common control.

2.3 Terrain branch and history feedback

Terrain base is not ordinary Full. It fits a shared history drift B_0 then independently fits a residual conditioner and diffusion for each of ten terrain configurations:

$$dX_t = \{B_0\phi_{\text{hist}}(t) + C_g\tilde{f}_g(X_t, t)\}\,dt + L_g\,dW_t, \qquad L_gL_g^\top = Q_g. \tag{2}$$

There are 404 training and 81 validation transitions, using each window's first future transition with equal row weight. The first ridge minimizes an unnormalised squared sum with $\rho = 10^{-6}$; the correction minimizes mean loss with $\lambda = 10^{-4}$ (normal-equation penalty $404\lambda = 0.0404$). Neither is joint optimisation. Terrain $Q_g = n^{-1}\sum_i \Delta t_i\zeta_i\zeta_i^\top$ is an uncentred rate-residual second moment, without a degrees-of-freedom correction, and is not the ordinary method's R_m.

The feature groups are history, retained-road subset, river, cover and surface, including road–cover and surface–history interactions. Road/river direction points towards the nearest feature point, not along its tangent or flow. The road filter excludes bridleway, cycleway, footway, path,

Table 2: Scientific interpretation of registered procedure comparisons. The original five families, 28 identities and 21 primary contrasts are kept; identities are not counts of independent algorithms.

Comparison	What changes; what must not be attributed
Pointwise / single / explicit	Mixing or representation; explicit also changes mode count. Not an isolated velocity-feature effect.
QMLE vs Full	Saved Θ, R, π are identical. Finite random score differences do not show a training-objective effect.
Mixed vs pure-ES	Same saved parameters; both add validation drift fitting relative to Full. Not an isolated loss-function comparison.
Scratch / Reptile / drift-only	Implemented fit/blend procedures. Mode indices are not proven semantically aligned; two-step is not a new sequential optimizer [12].
Observation interval	Fit data, noise binding $Q = \tau R$ and history/mode cadence change together; maximum integration h remains 5 s.
EM / Euler / d2	EM and Euler are update aliases. Two d2 score routes share predictions and common scoring; they do not independently validate their native estimators.
MC / CRN	Propagation procedures with their named Full20 control, not generic comparisons against an interchangeable pooled Full.

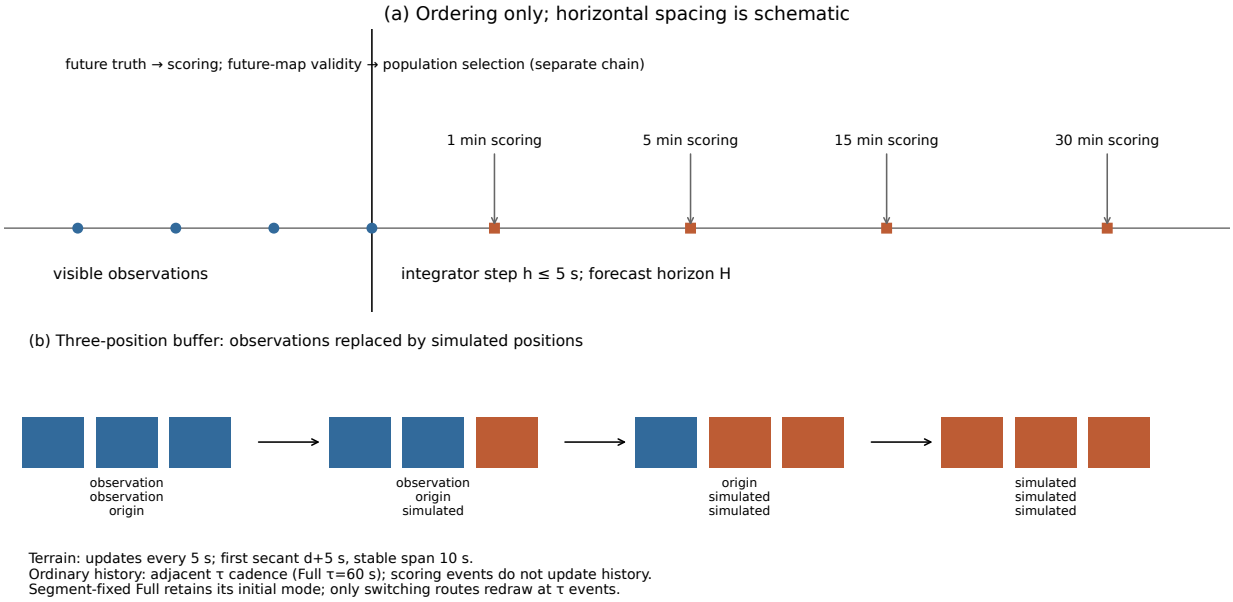
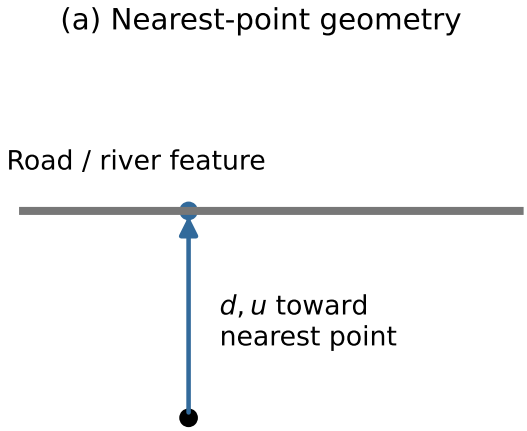
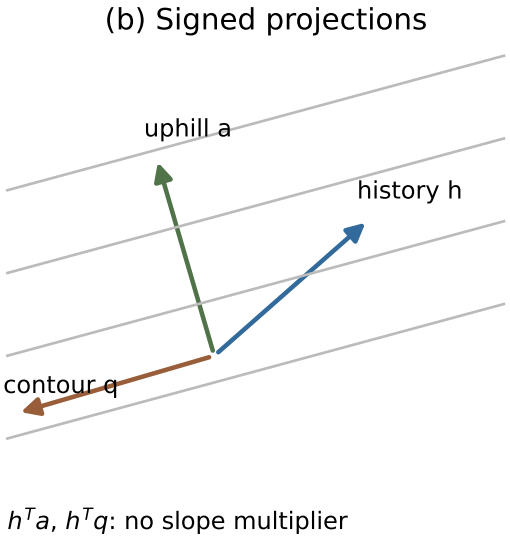


Figure 1: Observed prefix and simulated history. τ is the observation/model cadence, h the integration step and H the forecast horizon. Terrain updates its three-position buffer on 5 s events; once entirely simulated, its secant spans 10 s. Ordinary Full has a separate $\tau = 60$ s cadence. Future observations enter scoring only; scoring an event does not itself update history.

pedestrian, steps and track; it is not the complete pedestrian network. Surface features separate the signed history, uphill and contour directions; slope does not multiply their directional dot products. Figure 2 defines these inputs, and Figure 3 shows which are retained.





(c) Frozen WorldCover groups and road interaction

built	vegetation	water/wetland	valid other
50	10,20,30,40, 95,100	80,90	other valid codes

$1_{\text{group}} \log(1 + d/(1\,m))$; nodata is missing, not valid other.

Figure 2: Input geometry (three schematic panels, no measured effect). Distance and direction point toward the nearest feature; signed uphill and contour projections do not include a slope multiplier. Slope magnitude is a separate input, not depicted here. The frozen WorldCover groups are built (50), vegetation (10/20/30/40/95/100), water/wetland (80/90), and other valid codes. Invalid or nodata values are missing, not "other".

The overall comparison is all-terrain minus base. Leave-one-out (LOO) compares all-terrain with all-minus-group; leave-one-in (LIO) compares history-plus-group with base. A negative candidate-minus-control score favours the candidate. Describing a removal loss as factor benefit reverses the sign and maps an interval $[l, u]$ to $[-u, -l]$. The dependent columns in Figure 3 explain why removing one group can remove several numerical features. This design description remains useful even when required failures prevent the registered terrain inference.

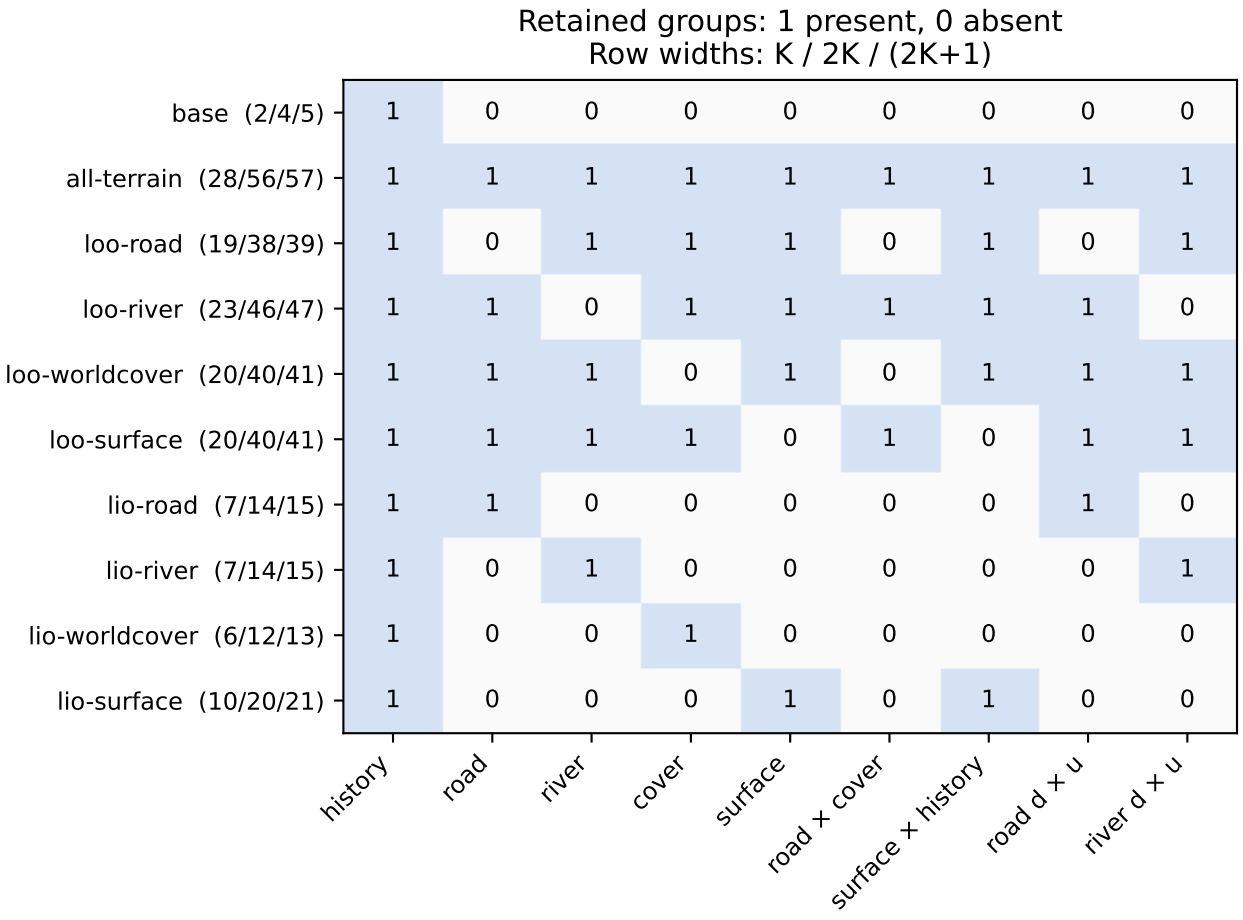


Figure 3: Ten configurations by nine group/dependency columns: 90 design states, not measurements. K counts numerical features, $2K$ includes their masks and $2K + 1$ includes the intercept; these are not fitted rank or degrees of freedom. Every row retains history. Road–cover requires both parent groups and surface–history remains in LIO-surface.

Saved model-source declarations identify WorldCover 2021 v200 [16], Overture 2026-08-19.0 transportation [23, 17], HydroRIVERS v1.0 [18, 19], and the retained DEM product/fallback rules for Mapzen Skadi and Copernicus GLO-30 [20, 21, 22]. Skadi is a mixed-source distribution, not certified pure NASA SRTM for every tile. Model sampling and case-display sources have separate records; product documentation does not certify tile acquisition epochs, map/track contemporaneity or route-publication permission.

Training/validation selected masks are all one, while future simulated queries can be invalid. The code zero-fills invalid numerical features and adds masks; robustness to such out-of-training-distribution inputs has not been learned or demonstrated. With penalised intercept and K constant-one masks, the minimum constant-subspace penalty is

$$\min_{\sum_{j=0}^{K} a_j = c} \frac{\lambda}{2} \sum_{j=0}^{K} a_j^2 = \frac{\lambda c^2}{2(K+1)}. \tag{3}$$

Base has $K = 2$ and all-terrain $K = 28$, so this denominator changes from 3 to 29. This is not a 9.7-fold change of the entire model's penalty. Unstandardised scales, redundant columns, independent refitting and diffusion changes further confound pure information attribution. Even complete forecasts would establish configuration increments, not isolated terrain effects.

3 Data, targets and evaluation

3.1 A retrospectively selected population

The released data contain 12,370 windows from 1,094 recording-hash blocks. Temporal eligibility leaves 127 windows/89 blocks; joint feature validity leaves 106/73; the fixed primary-origin selection leaves 46/46. The 130 windows having at least 1,800 s follow-up are not identical to the 127 passing all temporal criteria. Among temporally eligible windows, 21 fail joint map support. The 27 eligible but unselected blocks and extra windows are selection outcomes, not forecast failures.

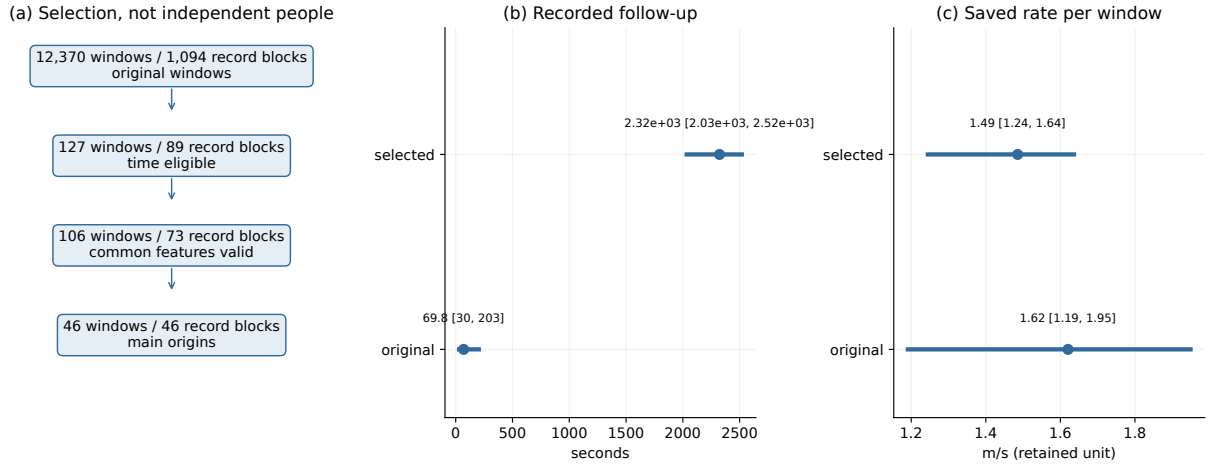


Figure 4: Window and recording-block funnel, with released versus selected follow-up duration and saved speed median/IQR. These are population descriptions, not predictor estimates. Eligibility tests map validity along the observed future route; it cannot be determined at deployment from the prefix alone.

Future true positions are not inputs to prediction: online queries use simulated positions. Nevertheless, future-route support is used to select this evaluation population, an important post-hoc restriction. The selected cohort spans 12 countries and 27 source labels; 33/46 blocks are from Great Britain, Belgium or Luxembourg. It is neither a globally representative sample nor a deployment-qualified cohort. The evaluation data were used previously; the complete influence of earlier inspection on design choices is not recoverable. It is not a fresh blinded holdout.

Exact saved-identity and full-sequence checks do not identify participants, nearby routes, repeated visits or formatting-modified duplicates. The block resampling therefore depends on unverified independence/exchangeability assumptions. Six known-velocity and six point-only cases reuse the primary records: there are 46 unique recording blocks and 58 origin–mode cases, not 58 participants or independent blocks.

3.2 Saved-clock horizons and metric definitions

Forecasts stop at the actual matched target, rather than interpolating an observation to an exact nominal horizon. Nominal 1/5/15/30 minute targets have retained elapsed-time ranges 49–67/294–305/889–910/1785–1814 s. Relative time reconstructed in the R1 data procedure and the condition tick used in prediction are separate objects. Exact tick joins and datetime precision do not authenticate physical acquisition time or UTC. An absolute offset affects solar/calendar interpretation; a relative-scale distortion affects rates, diffusion units and the meaning of a physical 30 minutes. No timezone correction is guessed. Results throughout are for the saved clock's nominal horizons; S is a calendar proxy, not a validated daylight mechanism. Source reconciliation is in the audit record.

For particles $x_1, \ldots, x_N$ and target y at a matched time, the execution-bound finite-ensemble score is the U-statistic, for $N \geq 2$:

$$\mathrm{ES}_N = \frac{1}{N} \sum_i \|x_i - y\| - \frac{1}{2N(N-1)} \sum_{i \neq j} \|x_i - x_j\|. \tag{4}$$

The bound scorer sums ordered particle pairs; diagonal distances are zero. Its denominator is $2N(N-1)$, not the empirical-distribution $2N^2$ form incorrectly printed in the previous main text. This corrects the description, not the saved scores or decisions. For the deterministic inertial reference, ES is evaluated directly as $\|x - y\|$, without applying the $N \geq 2$ formula to one particle. ES has units of metres; lower is better. Weighted ES is the equally weighted average over the four targets. Mean-position error is $e_j = \|\bar{x}_j - y_j\|$, with $\mathrm{ADE} = \frac{1}{4}\sum_{j=1}^{4} e_j$ and $\mathrm{FDE} = e_4$. ADE is not a dense-path integral; FDE is neither best-particle error nor mean particle error. Marginal disks are centred on the ensemble mean, with radii from particle-distance quantiles. Coverage and area are evaluated at each horizon, not as joint path regions. Average area is $\mathrm{E}[\pi r^2]$, not $\pi \mathrm{E}[r]^2$. The radius is the $\lceil qN \rceil$-th ordered particle distance at nominal level q. Enclosing approximately 90% of simulated particles does not guarantee 90% coverage of real targets. For a multimodal cloud the mean may lie in a low-density gap and the disk may include inaccessible terrain; it is neither a highest-density region nor a route-constrained forecast.

The completed inventory is $28 \times 58 \times 5 = 8120$ method forecasts, $10 \times 58 \times 5 = 2900$ terrain forecasts, 290 same-grid replay forecasts and 58 inertial paths: 11,368 score slots. The registered scientific 11,020 contain 11,015 successes and five failures; all score slots contain 11,363 successes and those same five failures. A score-slot count is not an array audit scope or number of independent observations.

3.3 Conditional paired inference

Within each recording block, the five fixed seeds are averaged before paired block inference. They quantify simulation variation, not independent people or training uncertainty. Original family-wise simultaneous bootstrap-t intervals use 2,000 draws per stream and shared within-family block indices; Holm-adjusted zero tests [13] have a separate role. The practical margin is $\delta = 41.100259$ m, 5% of validation inertial ES, not a validated search-and-rescue utility threshold. Planning used only three development blocks. A five-contrast family's single tail has about ten expected draws at probability 0.005; a four-contrast family's tail has about 12.5. These are finite precision limitations, not a simultaneous coverage theorem [14].

All deltas are candidate minus named control; negative ES is favourable. An interval crossing zero is not evidence of equivalence. Practical benefit or harm requires the entire original interval beyond $-\delta$ or $+\delta$ and its original statistical, seed and qualification gates. Equivalence, inconclusive information, unavailable evidence and failed program qualification are different states. Zero-SD d2 aliases do not independently demonstrate power; the dt600 comparison has low planning power. Full procedure and precise endpoints remain in the scientific supplement.

4 Results

4.1 Simple extrapolation first: the horizon reversal

On the same 46 primary blocks and original targets, saved weighted ES is 477.85 m for Full and 824.73 m for inertia. Their stored difference rounds to -346.87 m, computed from unrounded values, not subtraction of displayed means. Inertia is better at 1 and 5 minutes; Full is better at 15 and 30 minutes (Figure 5). The four isolated target times do not locate an exact crossover time. This descriptive reference comparison is not a newly registered superiority test.

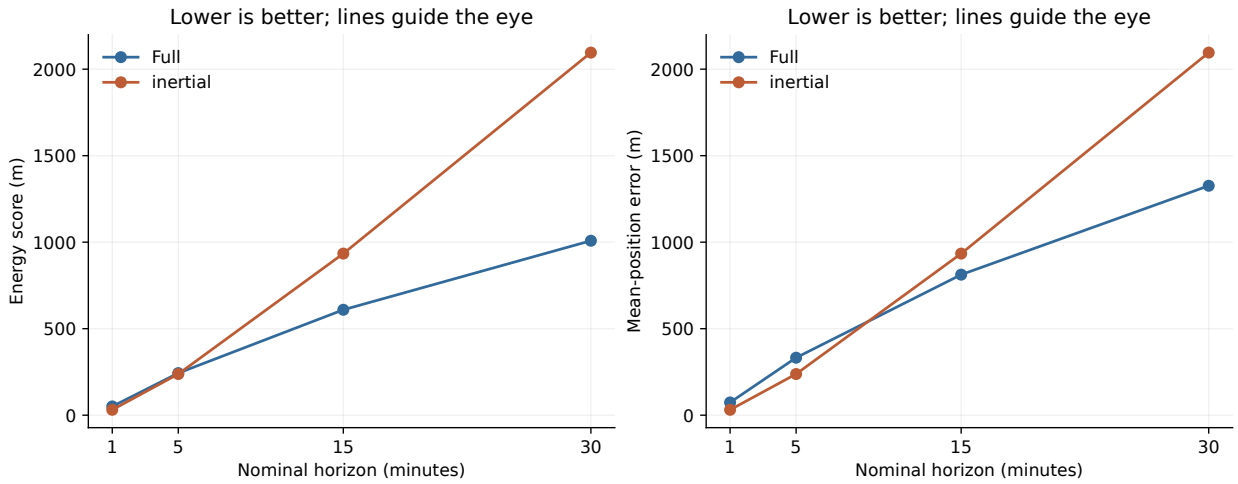


Figure 5: Full versus deterministic inertia at four nominal horizons. Left: ES; right: error of the predictive mean. The curves use the original equal-block/equal-seed summaries. Lines guide the eye, not interpolation or evidence for a precise crossover.

The deterministic reference is a valid point-mass distribution for ES, but does not test whether this stochastic model is necessary. A matched probabilistic constant-velocity state-space baseline is a separate future experiment, not a result presented here.

Table 3: Saved error of the predictive mean (m), with original target matches and four-model scope. These are not ES or mean particle errors.

Model	1 min	5 min	15 min	30 min
Full	73.77	332.31	812.26	1326.09
GMM	71.92	323.79	792.99	1299.11
dt300	75.88	340.10	825.76	1335.73
Inertial	31.17	237.93	934.03	2095.78

4.2 A better distribution score need not locate the mean better

The 300 s observation-interval procedure illustrates the distinction. Relative to its named Full01 control, weighted ES changes by -23.40 m but FDE by $+9.64$ m. Its mean-position error is also larger at each of the four saved targets (Table 3). ES accounts both for distances to the target and for spread between particles; it cannot be converted into mean-position error. The observation-interval comparison changes training exposure and diffusion binding as well as cadence, so this contrast does not isolate a numerical integration effect.

The dt300 ES simultaneous interval is $[-48.27, -2.47]$ m. Excluding zero does not satisfy the registered practical-benefit criterion: the whole interval would have to be below -41.100259 m, with the other gates met. The point thus communicates a metric conflict, not a practically superior method. All differences are calculated from unrounded source values.

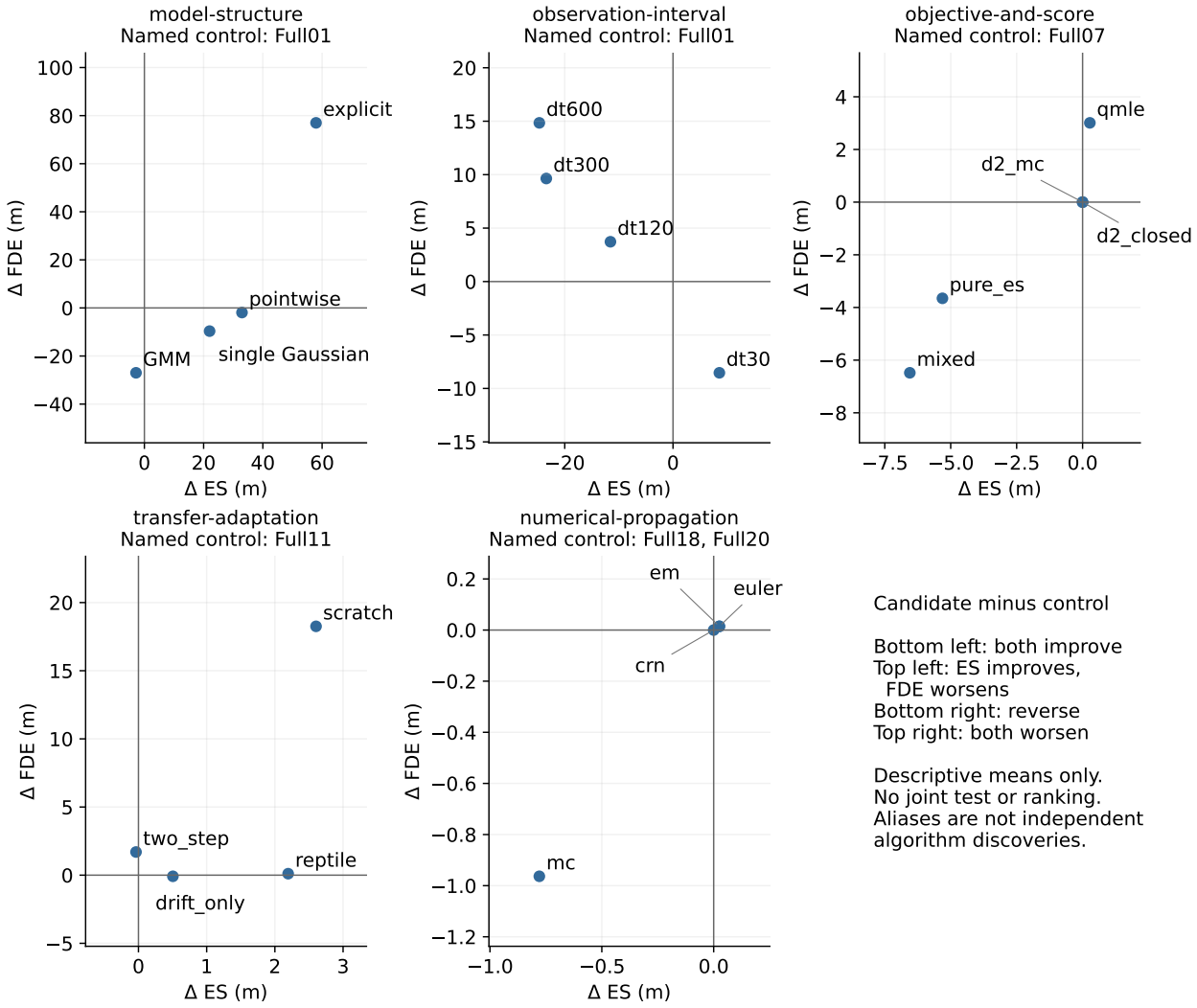


Figure 6: Distribution-score versus endpoint-error trade-offs. All 21 comparisons (42 displayed scalar differences) retain five registered families and named controls. Each point averages five simulation seeds within each of 46 paired blocks, then weights blocks equally; ES first averages four targets while FDE uses only the last. Negative is favourable on both axes; the upper-left quadrant improves ES but worsens FDE. Facets have different scales. These means have no joint confidence region and do not establish a Pareto or global algorithm ranking.

4.3 Long-horizon marginal disks are undercovered

Full's nominal 90% coverage is 95.65%, 37.83%, 35.22% and 46.52% at the four targets. At one minute, the 50% disk covers 33.04% while the 90% disk covers 95.65%; at thirty minutes even the 95% disk covers only 52.17%. Thus a single demonstrated radius rescaling is not the explanation. Across all 28 identities, thirty-minute 90% coverage is 21.74–79.13%. The full ES/area/coverage-

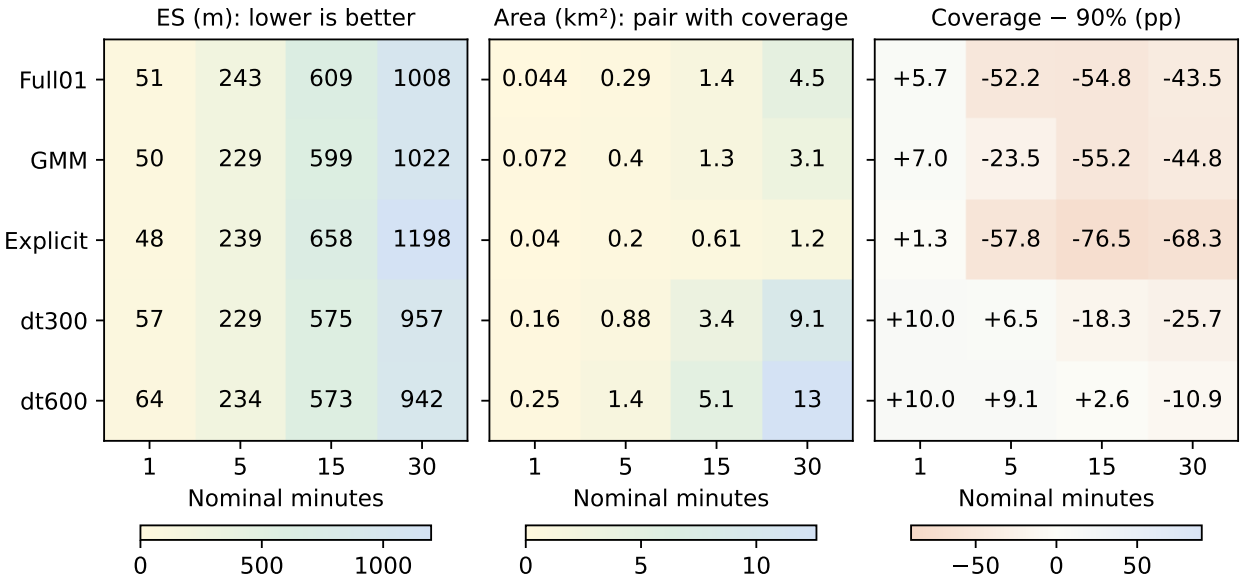


Figure 7: Role-selected overview: reference, residual mixture, explicit representation and long observation intervals. Rows are fixed by scientific role, not performance rank. ES and area use the full 28-identity ranges; coverage deviation is observed minus 90% in percentage points. The 60 cells are five roles × four horizons × three metrics, each averaging five seeds within each of 46 blocks, then blocks equally. All numbers are black on pale colours; equal positive/negative departures have equal colour distance from zero. Area is size, not a standalone quality score.

deviation overview is in the scientific supplement; its diverging colour scale is centred on nominal coverage, so overcoverage is not visually rewarded as "better".

Full's thirty-minute mean 90% disk area is 4.534 km^2 and mean radius 1183.6 m; median block area is a different 4.022 km^2 summary. The 300 s interval procedure has 9.109 km^2 mean area and 64.35% coverage, with FDE 1335.73 m; the 600 s procedure has 12.609 km^2, 79.13% and 1340.94 m. Larger regions cover more but do not make the ensemble mean more accurate or establish conditional calibration. No radius is retuned after inspecting this evaluation. The dt300 disk is

Table 4: Thirty-minute nominal 90% disks: three configurations and nine saved summary values. Each configuration uses 46 blocks and five ensembles per block. Radius and area are averaged separately; the radius of the mean area is not the mean radius.

Model	Coverage (%)	Mean area (km^2)	Mean radius (m)
Full01	46.52	4.534	1183.6
GMM	45.22	3.079	989.5
dt300	64.35	9.109	1692.8

about twice Full's mean area, yet its inclusion rate remains below the nominal 90%. A larger region

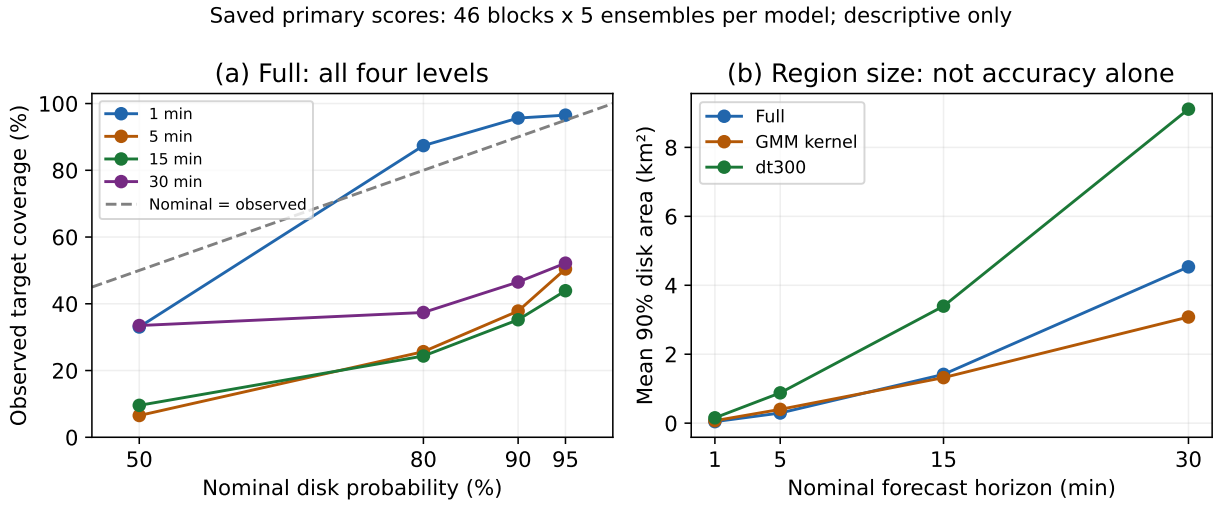


Figure 8: Calibration and size from the saved 46-block, five-seed summaries. Left: 16 Full coverage values (four probability levels at four horizons); closeness to the diagonal, not the highest curve, is the target. Right: 12 mean areas (three configurations at four horizons), to be read with coverage rather than ranked alone. Connecting lines guide reading; they do not add intermediate horizons or a calibration test.

can include more targets while its centre is less accurate. Conversely, GMM's smaller region does not by itself establish better uncertainty prediction. Coverage averages the same real target across five simulated disks in each block; there are 46 targets per horizon, not 230 independent targets. At block level this five-seed inclusion proportion takes only 0, 20, 40, 60, 80 or 100%. The 552 points in the supplementary block plot can overlap at those six levels; neither a block median of 100% nor a dense-looking cluster is a conditional calibration certificate. These observations support reporting size and inclusion together while retaining the complete block summaries.

4.4 Component comparisons and conflicting metrics

None of the 21 original simultaneous intervals lies wholly beyond the registered $\pm\delta$ practical boundaries. The audited finite-algorithm states are 14 practically equivalent and seven inconclusive; all resolution-invariant states remain inconclusive. Human acceptance is not implied. The pointwise mixture raises ES by 32.95 m relative to Full01; the GMM difference is -2.83 m with an interval spanning zero. At 5 and 15 minutes GMM's descriptive ES advantage reverses at 30 minutes.

An ES improvement can accompany a larger endpoint error. The supplement's family-faceted ΔES–ΔFDE quadrants retain the separate Full controls rather than joining two unrelated rankings. Seed/block plots also distinguish two uncertainty sources: single-Gaussian seed deltas are all positive but its block interval is $[-14.19, 60.81]$ m; GMM's five are negative but its interval is $[-21.22, 16.33]$ m; dt600's five are negative with $[-67.65, 9.08]$ m. Seed-direction agreement does not settle population uncertainty. Full five-seed tables, CRPS, endpoint particle-error quantiles and all 120 original metric views remain available in the supplement.

4.5 Terrain evidence and failure-dependent availability

The five retained failures have different family consequences. F01 (known-velocity dt30) invalidates that mode's observation-interval family; F02 (known-velocity single Gaussian) its model-structure

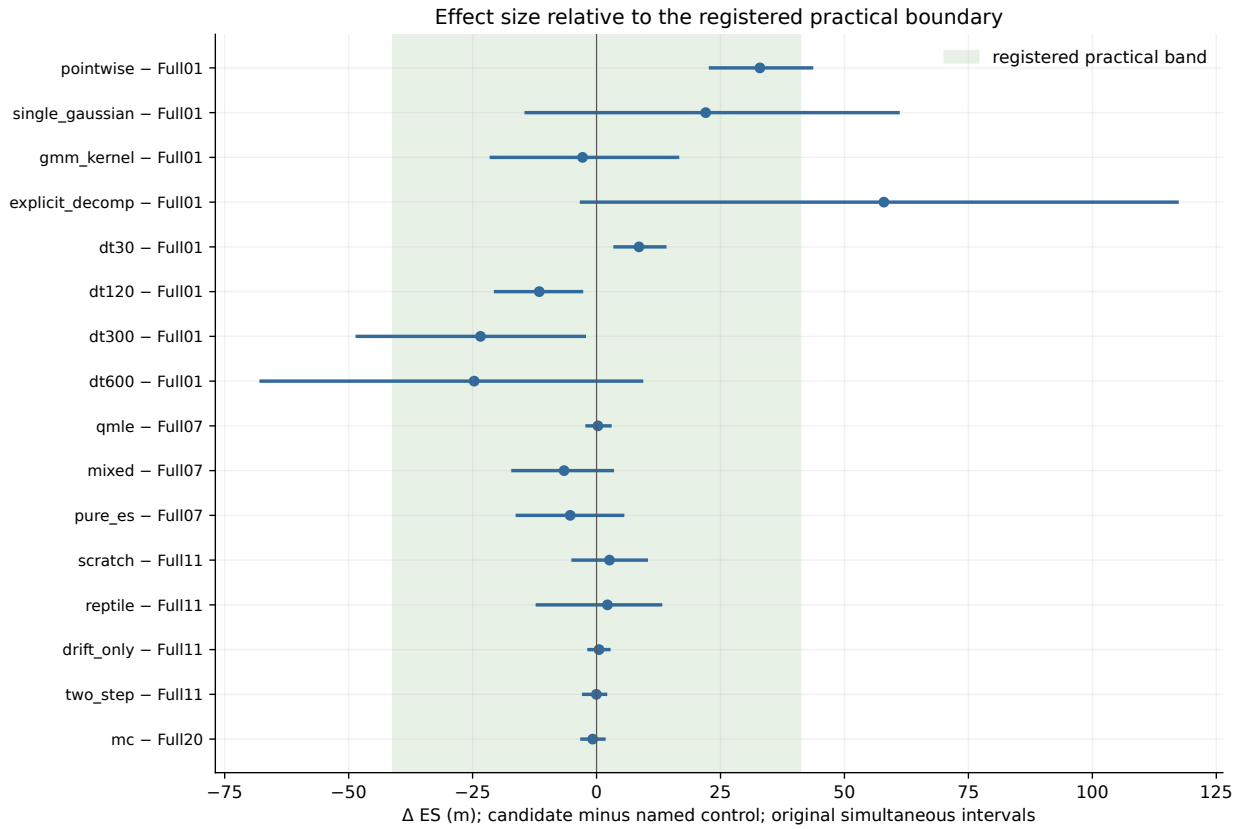


Figure 9: Original paired ES effects and simultaneous intervals, preserving each named candidate/control and the ±41.10 m practical band. Tiny implementation-consistency contrasts are separately displayed at their own scale in the supplement; they are not discarded or treated as scientifically large effects. Qualification and interval location are distinct.

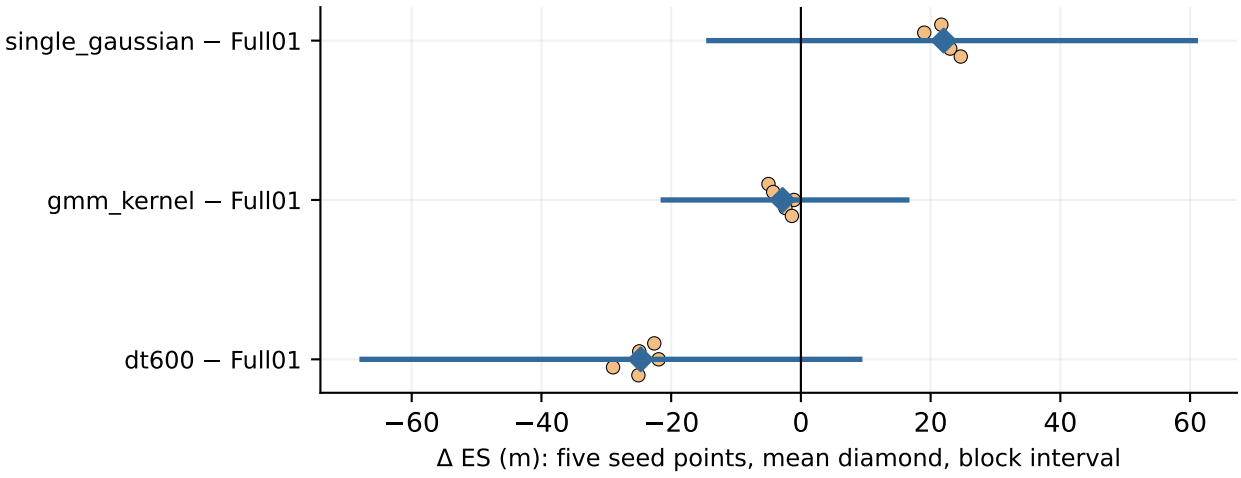


Figure 10: Three explanatory comparisons: 15 seed effects, three overall means and three original block simultaneous intervals. Each seed point averages 46 paired blocks; the interval resamples blocks after averaging seeds. The examples illustrate distinct uncertainty sources, without defining a new testing family or repeating model fitting.

family. F04 (primary all-terrain) invalidates the entire overall/LOO terrain family; F03/F05 (primary LIO-river/LIO-road) invalidate the entire LIO family. The overall/LOO grid has 1379/1380 successes and one failure; LIO has 1148/1150 and two failures, with no missing pending slots. These grids share baselines and must not be added as distinct forecasts.

The original complete-family rule does not permit a successful-subset factor ranking. "Unavailable" is not "no benefit", "zero effect" or "equivalent". Completed auxiliary tasks do not remove a retained failure. Five primary method families are complete but still have their own qualification states. The scientific supplement distinguishes family/mode availability; the audit record retains each failure identity and known logging limits. Missing exception stacks preclude a causal assertion of divergence, timeout, out-of-memory or a physical power loss.

4.6 Development diagnostics and computational cost

All-terrain's improved training fit with slightly worse validation fit is a development diagnostic, not a prediction benefit. The supplement plots all ten fit MSEs beside raw design columns/rank, separating rank deficiency from the successful augmented ridge solve. Model fit, positive covariance and numerical execution do not guarantee calibrated prediction. For example, the base/all-terrain training displacement-rate MSEs are 0.145796/0.136666 m^2/s^2, whereas validation MSEs are 0.077077/0.079117 m^2/s^2. These are 404 training and 81 validation transitions, not four-horizon position errors. The validation role differs from training, so its lower absolute MSE does not diagnose absence of overfitting. Nor does the slight increase identify overfitting as the cause. Constant masks, feature scales and refitted diffusion prevent a pure information-gain interpretation; Supplement Tables 40–41 retain all ten configurations and distinguish $2K + 1$ raw columns from design rank.

The original cost experiment contains 15 subjects, five provider-cold trials, one excluded prewarm and five provider-warm trials each: 165 trials, with 150 included in latency summaries. The supplementary plot reports p50/p95 latency and failures from these records, not the earlier isolated timings. Provider-cold does not flush the OS cache or represent cold deployment; five trials do not establish a stable population p95 or a speed–accuracy Pareto frontier. Map-query time is included in rollout, not added twice. Recorded interruption-exposed timing remains included and flagged, with no subtraction, exclusion or rerun. Memory samples and process-lifetime peaks do not identify exact per-trial peak memory.

5 Interpretation, limitations and conclusion

The primary empirical lesson is horizon-specific conflict among score, point accuracy and coverage. These predictors cannot be summarized by one winner or one uncertainty-scale adjustment. The calibration proxy's target and GMM labels differ from those of the evaluated predictive distribution; history cadence changes from observed to simulated support; missing-map inputs differ from development inputs. Each is a plausible mechanism, not an identified cause of the observed long-horizon undercoverage.

Attribution is additionally restricted by identical-parameter aliases, unmatched mode indices, multi-factor procedure changes and representation- dependent ridge penalties. Detailed execution accounting cannot repair these design limits. Nor can successful aggregate replay establish participant independence, physical UTC, generalization to unsupported routes or stability under training-data variation. Results condition on saved fits, selected records, fixed finite simulation and the resampling assumptions stated above.

5.1 Real-route illustrations and public boundary

Three existing base/all-terrain cases and their four-horizon route maps are restricted to the owner's local review manuscript. This public document does not contain those maps or case-level diagnostic files. Their original order, seed and forecasts cannot justify a terrain effect or population frequency. Route permission and map-layer redistribution are separate checks; neither authorship nor access to a local file grants publication permission. Author names, affiliations and declarations remain unconfirmed, not inferred from repository identities.

Future work should use a task-matched probabilistic constant-velocity baseline with the same prefix and targets, parameters chosen only on development data; test mode matching separately from parameter interpolation; calibrate the actual multi-step distribution with consistent component labels; match effective penalties in terrain comparisons; and validate relative/absolute clock provenance and independent-route/source grouping. These are explicit new designs, not experiments executed by this revision or prerequisites silently claimed complete.

RQ1 supports limited finite-procedure comparisons but no established practically beneficial/harmful component under the registered margin. RQ2 remains unavailable under its complete-family rule. The positive contribution is a reproducible, bounded account of short/long-horizon reversal, metric trade-offs and severe marginal undercoverage. It is not a validated rescue system, a general terrain ranking or proof of a new superior algorithm.

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